

Lexical Mapping and MML Tracking

Analysis of 8,747 Thesaurus Terms

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Introduction & Research Goals

What are we doing?

This research evaluates the structural alignment between a **Conceptual Metaphor Thesaurus** and the **WordNet** lexical database. We aim to see if metaphorical logic is reflected in linguistic hierarchy.

- ▶ **Lexical Mapping:** Linking 8,747 thesaurus headwords to specific WordNet synsets.
- ▶ **MML Tracking:** Analyzing "Hypernym Chains" to find keywords from the Metaphorical Mapping Logic (MML) (e.g., *QUALITY IS MONEY*).
- ▶ **Validation:** Determining if the "Source" and "Target" of a metaphor actually appear in the word's dictionary ancestry.

Mapping Success and Gaps

Thesaurus Statistics

- ▶ **Total Words in Thesaurus:**

8,747

- ▶ **Words Not Found:** 1,006

- ▶ **Words Found/Mapped:** 7,741

Unmapped Rate

11.5%

1,006 words failed to return a result. This "Lexical Gap" identifies terms that are either multi-word expressions or absent from the OMW 1.4 database.

Total Data Breakdown (8,747 Words)

- ▶ **Total Thesaurus Entries:** 8,747
- ▶ **Words Successfully Mapped:** 7,741 (88.5%)

The "3,072" Discrepancy

The mapping identifies **4,669 unique lemmas**. These lemmas appear a total of **7,741 times** across the thesaurus. The remaining **3,072** entries are duplicate instances of these lemmas in different conceptual categories (e.g., "pay" appearing in multiple themes).

MML Tracking: Unique Lemmas vs. Total Instances

Tracking Status	Unique Lemmas (4,669 words)	Total Instances (7,741 entries)	MML Requirement
Both Hypernyms	641	561*	Matches Target AND Source.
One Hypernym	1,366	3,184*	Matches Target OR Source.
Mapped (No Track)	2,662	3,996*	Mapped but lineage is distinct.
Total Found	4,669	7,741	

**Note: Duplicate instances account for the distribution differences between vocabulary and total entries.*

Structural Tracking Examples

Example 1: Sapphire (COLOUR IS MINERAL)

- ▶ **Sense A:** "...chromatic **color**..." → **Matches: COLOUR**
 - ▶ **Sense B:** "...gemstone... **mineral**..." → **Matches: MINERAL**
 - ▶ *Result: Lemma tracks **BOTH** domains.*
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Example 2: Pay (QUALITY IS MONEY/WEALTH)

- ▶ **Sense A (remunerate):** "...a sum of **money** paid..." → **Matches: MONEY**
- ▶ **Sense B (be worth it):** "...have the **quality** of being..." → **Matches: QUALITY**
- ▶ *Result: Lemma tracks **BOTH** domains.*

Summary of Findings

- ▶ **High Coverage:** 88.5% of the thesaurus was successfully mapped to WordNet, showing strong lexical overlap.
- ▶ **Structural Evidence:** 641 unique lemmas show "Full Tracking," meaning the metaphor is explicitly encoded in the dictionary's hierarchy.
- ▶ **Lexical Ambiguity:** Many words (3,072 instances) function in multiple metaphorical domains, requiring sense-disambiguation to track correctly.
- ▶ **Conclusion:** WordNet hypernyms provide a viable, though partial, framework for automatically validating conceptual metaphors.

Final Tasks and Project Completion

Immediate Next Steps

To finalize this phase of the research, we are focusing on:

- ▶ **Manual Validation:** Reviewing a random sample of the "Mapped (No Track)" words to see if the MML keywords are present under different synonyms that the current script missed.
- ▶ **Refining the Lexical Gap:** Investigating the 1,006 "Not Found" words to determine if they are mostly multi-word expressions (e.g., "silver lining") or modern slang missing from WordNet.
- ▶ **Statistical Correlation:** Calculating the percentage of "Both/One" tracking across different metaphorical themes to see which metaphors are most "dictionary-stable."

Thank you for your attention!

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